

**PRAKTIKUM BASIS DATA**

**MODUL 12**

**NoSQL\_Columnar\_Database**



Oleh:

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**PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK**

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**UNIVERSITAS TELKOM**

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## JURNAL

1. Buatlah sebuah database Duckdb baru dengan nama jmmodul12.db. Kemudian, lakukan import data dari file jmmodul12.

Jawaban:

```
C:\Users\felix\Downloads\duckdb_cli-windows-amd64>duckdb jmmodul12.d
DuckDB v1.5.3 (Variegata)
Enter ".help" for usage hints.
jmmodul12 D .read jmmodul12.txt
jmmodul12 D SHOW TABLES;
```

| name           |
|----------------|
| varchar        |
| halte_busway   |
| tap_in_out_log |

2. Summarize

Jawaban:

```
jmmodul12 D SUMMARIZE tap_in_out_log;
```

| column_name      | column_type   | min                             | ... | q75                 | count | null_percentage |
|------------------|---------------|---------------------------------|-----|---------------------|-------|-----------------|
| varchar          | varchar       | varchar                         | ... | varchar             | int64 | decimal(9,2)    |
| id_transaksi     | UUID          | 1155c729-acfd-4c16-ac11-d511... | ... | NULL                | 30    | 0.00            |
| nomor_kartu      | VARCHAR       | K-001234                        | ... | NULL                | 30    | 0.00            |
| id_halte_asal    | VARCHAR       | H01                             | ... | NULL                | 30    | 0.00            |
| id_halte_tujuan  | VARCHAR       | H01                             | ... | NULL                | 30    | 0.00            |
| waktu_tap_in     | TIMESTAMP     | 2026-05-12 06:00:00             | ... | 2026-05-12 17:30:00 | 30    | 0.00            |
| waktu_tap_out    | TIMESTAMP     | 2026-05-12 06:45:00             | ... | 2026-05-12 18:40:00 | 30    | 0.00            |
| tarif_debet      | DECIMAL(10,2) | 3500.00                         | ... | 3500                | 30    | 0.00            |
| sisa_saldo_kartu | DECIMAL(12,2) | 2000.00                         | ... | 54000               | 30    | 0.00            |
| tipe_pembayaran  | VARCHAR       | BCA Flazz                       | ... | NULL                | 30    | 0.00            |
| jenis_perjalanan | VARCHAR       | Lansia                          | ... | NULL                | 30    | 0.00            |
| catatan_sistem   | VARCHAR       | Saldo Kritis                    | ... | NULL                | 30    | 0.00            |

11 rows use .last to show entire result 12 columns (6 shown)

1. Distinct tipe\_pembayaran

```
jmmodul12 D SELECT COUNT(DISTINCT tipe_pembayaran)
FROM tap_in_out_log;
```

| count(DISTINCT tipe_pembayaran) |
|---------------------------------|
| int64                           |
| 4                               |

2. Minimum sisa\_saldo\_kartu

```
jmmodul12 D SELECT MIN(sisa_saldo_kartu)
              FROM tap_in_out_log;
```

|                                        |
|----------------------------------------|
| min(sisa_saldo_kartu)<br>decimal(12,2) |
| 2000.00                                |

### 3. Maksimum tarif\_debet

```
jmmodul12 D SELECT MAX(tarif_debet)
              FROM tap_in_out_log;
```

|                                   |
|-----------------------------------|
| max(tarif_debet)<br>decimal(10,2) |
| 3500.00                           |

### 3. CSV

Jawaban:

#### a. Tampilkan 5 data pertama

```
jmmodul12 D SELECT *
              FROM 'halte_busway.csv'
              LIMIT 5;
```

| id_halte<br>varchar | nama_halte<br>varchar | koridor_utama<br>int64 | ... | alamat_lengkap<br>varchar | kecamatan<br>varchar | kota_administrasi<br>varchar |
|---------------------|-----------------------|------------------------|-----|---------------------------|----------------------|------------------------------|
| H01                 | Blok M                | 1                      | ... | Jl. Sultan Hasanuddin     | Kebayoran Baru       | Jakarta Selatan              |
| H02                 | Bundaran HI           | 1                      | ... | Jl. MH Thamrin            | Menteng              | Jakarta Pusat                |
| H03                 | Harmoni               | 1                      | ... | Jl. Gajah Mada            | Gambir               | Jakarta Pusat                |
| H04                 | Monas                 | 1                      | ... | Jl. Medan Merdeka         | Gambir               | Jakarta Pusat                |
| H05                 | Kampung Melayu        | 5                      | ... | Terminal Kp Melayu        | Jatinegara           | Jakarta Timur                |

5 rows use .last to show entire result 12 columns (6 shown)

#### b. Filter Jakarta Pusat dan Hitung jumlah halte Jakarta Pusat

```
jmmodul12 D SELECT *
              FROM 'halte_busway.csv'
              WHERE kota_administrasi = 'Jakarta Pusat';
```

| id_halte<br>varchar | nama_halte<br>varchar | koridor_utama<br>int64 | ... | alamat_lengkap<br>varchar | kecamatan<br>varchar | kota_administrasi<br>varchar |
|---------------------|-----------------------|------------------------|-----|---------------------------|----------------------|------------------------------|
| H02                 | Bundaran HI           | 1                      | ... | Jl. MH Thamrin            | Menteng              | Jakarta Pusat                |
| H03                 | Harmoni               | 1                      | ... | Jl. Gajah Mada            | Gambir               | Jakarta Pusat                |
| H04                 | Monas                 | 1                      | ... | Jl. Medan Merdeka         | Gambir               | Jakarta Pusat                |
| H07                 | Tosari                | 1                      | ... | Jl. Sudirman              | Menteng              | Jakarta Pusat                |
| H11                 | Pasar Baru            | 3                      | ... | Jl. Pos No. 1             | Sawah Besar          | Jakarta Pusat                |
| H16                 | Benhil                | 1                      | ... | Jl. Jend Sudirman         | Tanah Abang          | Jakarta Pusat                |
| H17                 | Senen                 | 2                      | ... | Jl. Stasiun Senen         | Senen                | Jakarta Pusat                |
| H24                 | Slipi Petamburan      | 9                      | ... | Jl. Letjen S. Parman      | Tanah Abang          | Jakarta Pusat                |
| H29                 | Sawah Besar           | 1                      | ... | Jl. Gajah Mada            | Sawah Besar          | Jakarta Pusat                |

```

jmmodul12 D SELECT COUNT(*) AS jumlah_halte
              FROM 'halte_busway.csv'
              WHERE kota_administrasi = 'Jakarta Pusat';

```

| jumlah_halte |
|--------------|
| int64        |
| 9            |

#### 4. Exclude

Jawaban:

```

jmmodul12 D SELECT *
              EXCLUDE (lokasi_lat, lokasi_lon)
              FROM halte_busway
              LIMIT 5;

```

| id_halte<br>varchar | nama_halte<br>varchar | koridor_utama<br>int32 | ... | alamat_lengkap<br>varchar | kecamatan<br>varchar | kota_administrasi<br>varchar |
|---------------------|-----------------------|------------------------|-----|---------------------------|----------------------|------------------------------|
| H01                 | Blok M                | 1                      | ... | Jl. Sultan Hasanuddin     | Kebayoran Baru       | Jakarta Selatan              |
| H02                 | Bundaran HI           | 1                      | ... | Jl. MH Thamrin            | Menteng              | Jakarta Pusat                |
| H03                 | Harmoni               | 1                      | ... | Jl. Gajah Mada            | Gambir               | Jakarta Pusat                |
| H04                 | Monas                 | 1                      | ... | Jl. Medan Merdeka         | Gambir               | Jakarta Pusat                |
| H05                 | Kampung Melayu        | 5                      | ... | Terminal Kp Melayu        | Jatinegara           | Jakarta Timur                |

5 rows

use .last to show entire result

10 columns (6 shown)

```

jmmodul12 D DESCRIBE halte_busway;

```

| halte_busway       |                  |
|--------------------|------------------|
| <u>id_halte</u>    | varchar not null |
| nama_halte         | varchar          |
| koridor_utama      | integer          |
| lokasi_lat         | double           |
| lokasi_lon         | double           |
| tipe_halte         | varchar          |
| fasilitas_toilet   | boolean          |
| fasilitas_mushola  | boolean          |
| status_operasional | varchar          |
| alamat_lengkap     | varchar          |
| kecamatan          | varchar          |
| kota_administrasi  | varchar          |

#### 5. Timer

Jawaban:

```

jmmodul12 D .timer on
jmmodul12 D SELECT h_asal.nama_halte,
                    h_tujuan.nama_halte,
                    COUNT(t.id_transaksi) AS jumlah
FROM tap_in_out_log t
JOIN halte_busway h_asal
  ON t.id_halte_asal = h_asal.id_halte
JOIN halte_busway h_tujuan
  ON t.id_halte_tujuan = h_tujuan.id_halte
GROUP BY h_asal.nama_halte,
         h_tujuan.nama_halte
ORDER BY jumlah DESC;

```

| nama_halte<br>varchar | nama_halte<br>varchar | jumlah<br>int64 |
|-----------------------|-----------------------|-----------------|
| Grogol 1              | Kampung Melayu        | 1               |
| Bundaran HI           | Kota                  | 1               |
| Pancoran Barat        | Slipi Petamburan      | 1               |
| Pinang Ranti          | Cawang UKI            | 1               |
| Pluit                 | Pasar Baru            | 1               |
| Ragunan               | Blok M                | 1               |
| Manggarai             | Blok M                | 1               |
| Ciledug               | Dukuh Atas 1          | 1               |
| Tosari                | Kuningan Timur        | 1               |
| Tanjung Priok         | Ciledug               | 1               |
| Kuningan Timur        | Bundaran HI           | 1               |
| Kuningan Timur        | Grogol 1              | 1               |
| Monas                 | Lebak Bulus           | 1               |
| Blok M                | Tosari                | 1               |
| Senen                 | Harmoni               | 1               |
| Harmoni               | Pulo Gadung           | 1               |
| Petamburan            | Grogol 1              | 1               |

|                    |                    |   |
|--------------------|--------------------|---|
| Slipi Petamburan   | Gatot Subroto LIPI | 1 |
| Pulo Gadung        | Pasar Baru         | 1 |
| Ancol              | Bundaran HI        | 1 |
| Gatot Subroto LIPI | Pancoran Barat     | 1 |
| Blok M             | Harmoni            | 1 |
| Cawang UKI         | Pinang Ranti       | 1 |
| Kota               | Tanjung Priok      | 1 |
| Kampung Melayu     | Tosari             | 1 |
| Matraman 1         | Senen              | 1 |
| Kota               | Ancol              | 1 |
| Lebak Bulus        | Dukuh Atas 1       | 1 |
| Bundaran HI        | Ancol              | 1 |
| Pasar Baru         | Pluit              | 1 |

30 rows

3 columns

Run Time (s): real 0.074 user 0.015625 sys 0.046875

jmmodul12 D .timer off